BackOnTrack - Focus Monitoring System

Project Documentation

1. Project Overview

BackOnTrack is a real-time focus monitoring system designed to analyze user engagement during computer-based work sessions. The system uses behavioral indicators such as eye movement, head orientation, and body posture to determine whether a user is attentive or distracted. Unlike traditional productivity tools that rely on fixed timers, this system evaluates physical attention signals in real time.

The project combines computer vision techniques with embedded hardware to create a hybrid monitoring system. A software based interface performs face tracking while a microcontroller monitors posture using motion sensors.

2. Problem Statement

Maintaining focus during long study or work sessions is a common challenge. Many users lose attention gradually without noticing it. Most available tools either use reminders or manual tracking methods, which do not consider actual behavioral indicators.

Key gaps identified:
• Lack of real-time behavioral monitoring
• No integration of posture with visual attention
• No automatic detection of distraction patterns

3. Objectives

The primary goal is to build a system capable of detecting whether a user is focused or distracted using measurable physical indicators.

Objectives include detecting eye and head movement, monitoring posture using IMU data, combining hardware and software inputs, generating focus statistics, and providing feedback alerts.

4. System Description

The system consists of a software interface responsible for facial tracking and a hardware unit responsible for posture monitoring and feedback generation. The software processes webcam

input while the ESP32 module reads posture data from the MPU6050 sensor. Both modules communicate through a shared data system and contribute to the final focus decision.

5. Hardware Components

Components	Purpose
Seeed XIAO ESP32C3	Microcontroller for processing
MPU6050 Module	6-Axis Gyro and Accelerometer sensor
Buzzer	Feedback Alert as beep
Battery	Power Supply
Switch	Switching on/off the circuit
LED	Power On Indicator
Webcam	Used for Image Processing

6. Software Technologies

Software	Purpose
MediaPipe	Face tracking
Chart.js	Visualization
Firebase	Data communication
JavaScript	Logic implementation
HTML/CSS	Interface design

7. Working Principle

The system analyzes both visual and motion data. The camera tracks facial landmarks while posture data is evaluated using the IMU. If both remain within acceptable limits the user is considered focused. The workflow includes face detection, eye analysis, posture reading, focus decision, timer update, and alert triggering.

8. Focus Detection Method

Focus detection uses threshold comparisons. Eye position determines gaze direction while head orientation confirms engagement. Posture detection uses roll and pitch values from IMU data. The final decision combines both inputs rather than relying on a single parameter.

9. Alert Mechanism

The system uses time filtering and triggers alerts only if distraction continues for a defined duration. Currently, a buzzer alert activates after 20 seconds of distraction and repeats after delay if distraction continues.

10. Data Tracking

The system tracks focus and distraction durations and displays them using a timer and pie chart. At session end a CSV report is generated containing session start time, end time, and focus statistics.

11. Challenges Faced & their solutions

- **Sensor noise in IMU readings**
  Handled by applying filtering techniques and complementary filter smoothing to stabilize readings.
- **Lighting affecting face detection**
  Improved by testing under different lighting conditions and adjusting detection thresholds.
- **False detection during movement**
  Reduced by introducing time-based validation so instant movements are ignored.
- **Camera angle dependency**
  Managed by defining acceptable detection ranges and recommending proper camera positioning.
- **Timing synchronization issues**
  Improved by optimizing data update intervals and separating sensor and communication tasks.
- **Stability during long operation**
  Addressed through continuous testing and improving timing logic for long sessions.

12. Limitations

The system's accuracy depends on proper lighting and correct camera placement. Since it detects physical signs like eye direction and posture, it cannot guarantee that a person is mentally focused. Detection may also become less accurate if the user's face is not clearly visible or if the head is turned too far away from the camera.

13. Applications

Student study monitoring, productivity tracking, and online learning environments are some potential applications of this system.

14. Conclusion

The BackOnTrack system demonstrates a practical real-time focus monitoring approach combining computer vision and embedded sensing. Further optimization and testing can improve reliability and usability.